

IMO(12th) – 1970

First Day

1. Given a point M on the side AB of the triangle ABC, let r_1 and r_2 be the radii of the inscribed circles of the triangles ACM and BCM respectively while ρ_1 and ρ_2 be the radii of the excircles of the triangles ACM and BCM at the sides AM and BM respectively. Let r and ρ denote the respective radii of the inscribed circle and excircle at the sides AB of the triangle ABC. Prove that:

$$\frac{r_1}{\rho_1} \frac{r_2}{\rho_2} = \frac{r}{\rho}.$$
2. Let a and b be the bases of the two number systems and let:

$$A_n = \overline{x_1 x_2 \dots x_n}^{(a)}, \quad A_{n+1} = \overline{x_0 x_1 x_2 \dots x_n}^{(a)}, \quad B_n = \overline{x_1 x_2 \dots x_n}^{(b)}, \quad B_{n+1} = \overline{x_0 x_1 x_2 \dots x_n}^{(b)}$$
 be numbers in the number system with respect bases a and b , so that $x_0, x_1, x_2, \dots, x_n$ denote digits in the number system with base a as well as in the number system with base b . Suppose that neither x_0 nor x_1 is zero. Prove that $a > b$ if and only if $\frac{A_n}{A_{n+1}} < \frac{B_n}{B_{n+1}}$.
3. Let $1 = a_0 \leq a_1 \leq a_2 \leq \dots \leq a_n \leq \dots$ be a sequence of real numbers. Consider the sequence $b_1, b_2, \dots$ defined by $b_n = \sum_{k=1}^n \left(1 - \frac{a_{k-1}}{a_k}\right) \frac{1}{\sqrt{a_k}}$. Prove that:
 - (a) For all natural numbers n , $0 \leq b_n < 2$.
 - (b) Given any arbitrary $0 \leq b < 2$, there is a sequence $a_0, a_1, a_2, \dots, a_n, \dots$ of the above type such that $b_n > b$ is true for an infinity of natural numbers n .

Second Day

4. For what natural number n can the product of some of the numbers $n, n+1, n+2, n+3, n+4, n+5$ be equal to the product of the remaining ones?
5. In the tetrahedron ABCD, the edges BD and CD are mutually perpendiculars, and the projection of the vertex D to the plane ABC is the intersection of the altitudes of the triangle ABC. Prove that $(AB + BC + CA)^2 \leq 6(DA^2 + DB^2 + DC^2)$. For which tetrahedron does equality hold?
6. Given 100 points in the plane, no three of which are on the same line, consider all triangles that have all their vertices chosen from the 100 given points. Prove that at most 70% of these triangles are acute angled.

 **